Foundational Player Auction Price Prediction Model Framework

Model Architecture

The proposed modelling framework consists of three distinct stages to produce price distributions reflecting the demand and supply side of player auction economics (Figure 1). This framework attempts to capture both the profile characteristics of a player, as well as, the stochastic nature of the auction environment. Data from previous IPL auctions will be used with the objective of creating a general price distribution model that can be replicated across multiple franchise leagues.

Dynamic player price

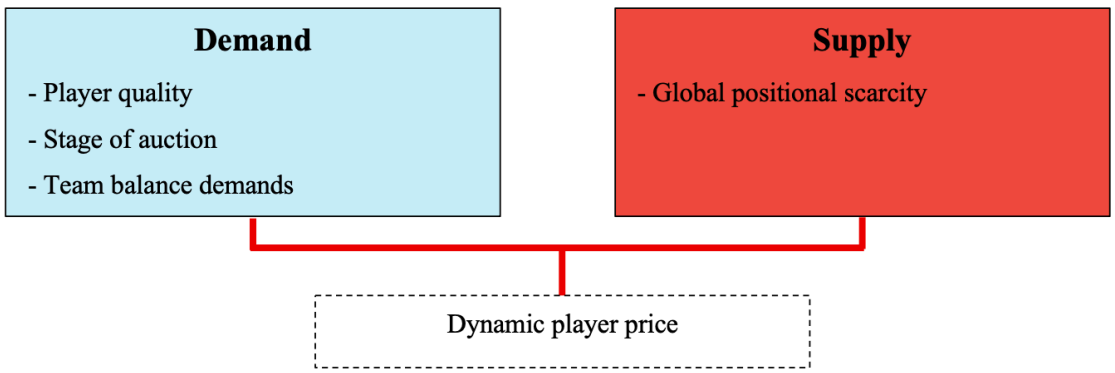

Figure 1: Demand and supply features driving dynamic player pricing

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Stage 1: Feature identification and engineering

The objective of this stage is to identify the features that are representative of a global market valuation signal with respect to a players profile characteristics, global positional scarcity, stage of the auction sold and base price. Correlations between proposed features and the proportion of team budget spend (the target value) will be produced using a series of regressions to test our assumptions on *what drives player pricing*. These features may include:

- Playing role (fast bowlers, spin bowlers, bowling all-rounders, top and middle-order batters)

Batting and bowling handedness

Nationality

Age

Matches played at the international level (across formats) and domestically (across various major T20 tournaments). This is used as a proxy for *player quality*.

Global positional scarcity (i.e., the proportion of players remaining in the auction with respect to *player role* and *underlying quality*, expressed as a %). Underlying quality in this case can simply be binned by analysts as high, medium and low.

Stage of auction player is sold

Base price

Other features identified by high performance staff (i.e., marketability)
A report with key findings will be produced at the conclusion of this stage.

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Stage 2: Random forest modelling and production of initial price distributions

Once key features have been identified, a Quantile Random Forest (QRF) model* will be trained to estimate the distribution of auction prices (converted from the proportion of team budget spend) given user-specified inputs. Price confidence is inferred directly from QRF outputs through the spread of the predicted distribution, with a tighter spread of prices indicating higher confidence in the estimated valuation. The resulting model can then be deployed to a *Streamlit* user interface (Figure 2) and used as a decision support tool to visualise the price distributions in different input scenarios. These distributions act as the foundational 'baseline estimates' that can be used in a Monte Carlo simulation.

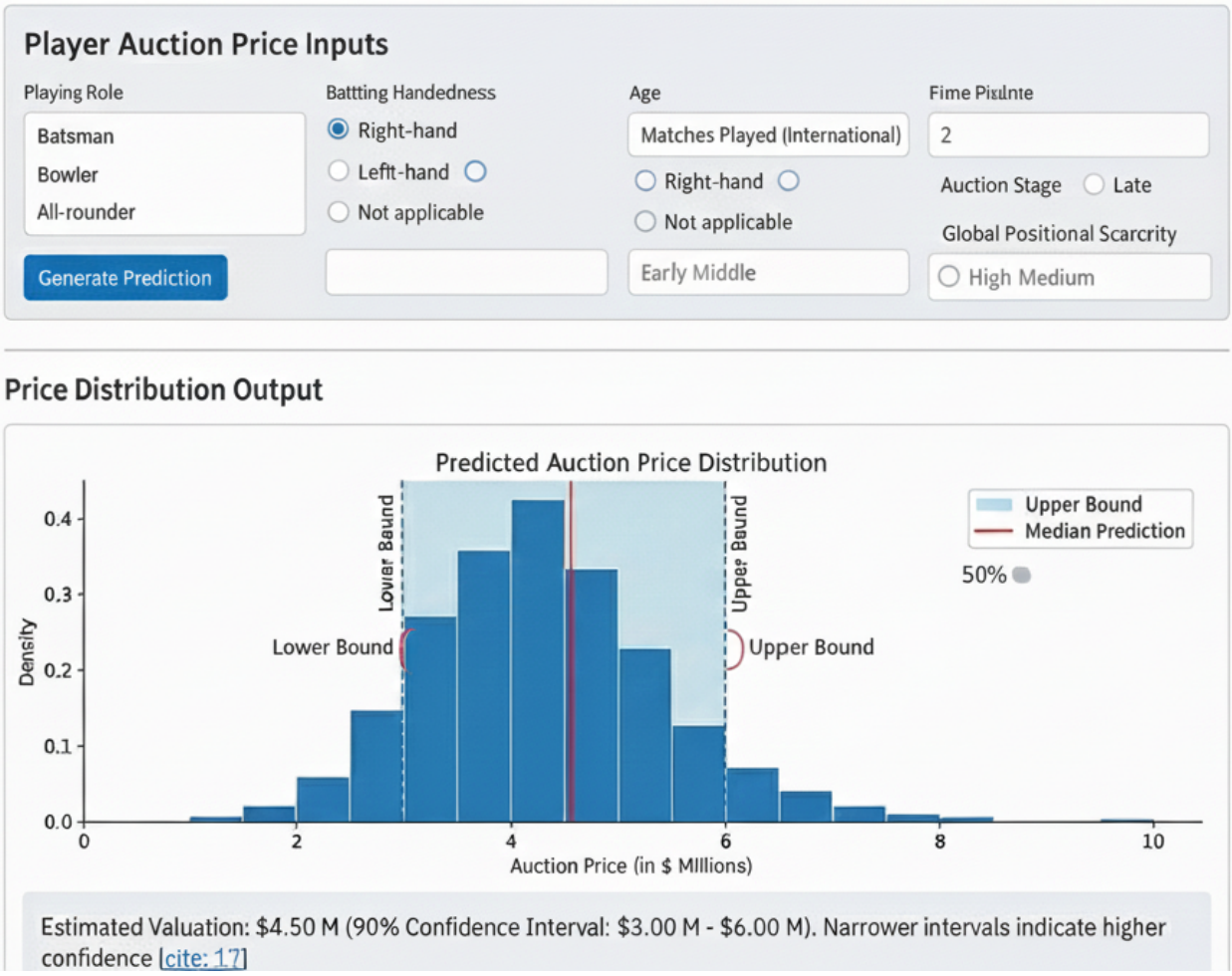

Figure 2: Auction dashboard that can produce initial baseline estimates of player price distributions

Stage 3: Monte Carlo simulation

To account for individual team behaviours in the auction environment that are unable to be captured in a machine learning model, a Monte Carlo simulation layer will be developed to capture stochastic fluctuations in demand arising from each team's requirement to:

(i) Satisfy minimum international and domestic player quotas

(ii) Construct a balanced starting twelve

(iii) Budgetary constraints (adjusted for players already retained)

A teams willingness to pay (WTP) is computed for each player within each simulation by randomly drawing a *prior price* produced by the QRF distribution to reflect inherent differences in player valuation between teams. This price is then adjusted by multipliers according to international and domestic player quotas, team balance requirements and budget constraints. Other factors such as historical bidding behaviours or proclivity to select former squad members can

considered in more advanced versions of the simulation architecture. The benefit of this approach is its scalability.

* A Quantile Random Forest (QRF) is a machine learning ensemble model that uses a collection of decision trees to estimate a range of possible outcomes rather than a single point prediction. Overfitting is mitigated through the random selection of

 features for each split and the use of bootstrapped training samples, which introduces diversity among trees and prevents any

single feature or data subset from dominating the model's predictions.

The following rules will govern the simulation:
1. *Teams will only place bids when the current bid price < remaining budget*.
2. *If all teams WTP values fall below the minimum base price, the player remains unsold*.
3. *Team player quotas, balance requirements and remaining budget are updated following each sale*.

By running the Monte Carlo simulation thousands of times, the model generates an updated distribution of player prices that incorporates dynamic auction interactions that can also be used in real time to inform in-auction decision-making. The dashboard in Figure 2 will be updated to incorporate this simulation architecture.

Interpretability & Usability
Both initial price (derived from QRF) and Monte Carlo derived distributions will be displayed on a graphical interface that can be accessed by coaches, analysts or auction strategists via the internet. Inputs can be modified using the fields displayed on this interface as illustrated in Figure 2.

Limitations & Trade-offs

To quantify *global positional scarcity*, all players are required to be categorised according to
quality (e.g., high, medium and low). This can be assessed subjectively by analysts or
quantified using rule-based decisions (e.g., if Player X > 50 T20i games THEN 'high').

Difficulty in determining the effect of multipliers to adjust the WTP score for each player.

Outlier auction prices can significantly skew distributions. These values will be excluded from modelling to prevent overfitting.

Confidence in model outputs will drop for rare player roles (i.e., LH Leg-spin) where there exists insufficient data.